

Semi-automated labelling for CV tasks with Paligemma

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Agenda

- Intro to Computer Vision(CV) and various CV tasks
- Vision Language Models (VLMs)
- Paligemma: Google's open model
 - Architecture details
 - Model capabilities
 - Live Demo
- Semi automated labelling with Paligemma
 - Live Demo

Section 01

Computer Vision and various vision tasks

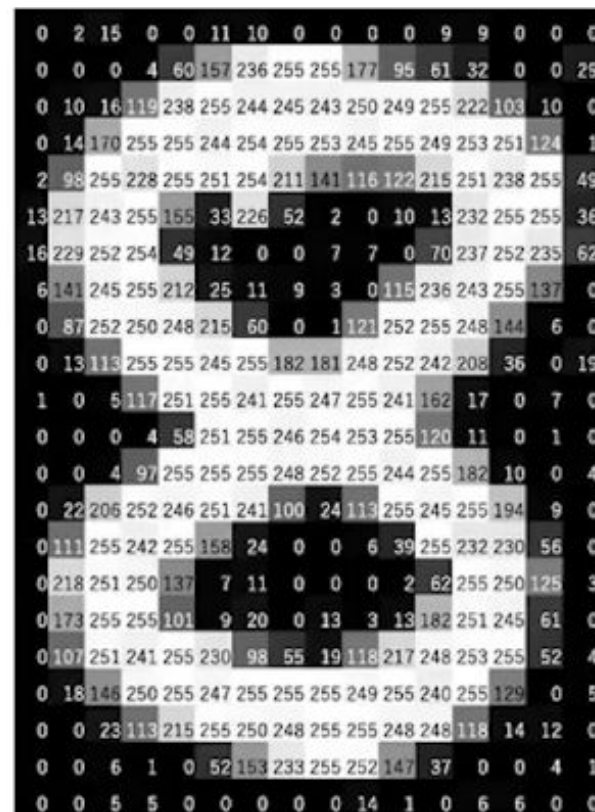
Section 1

Computer Vision

Computer vision is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images and videos.

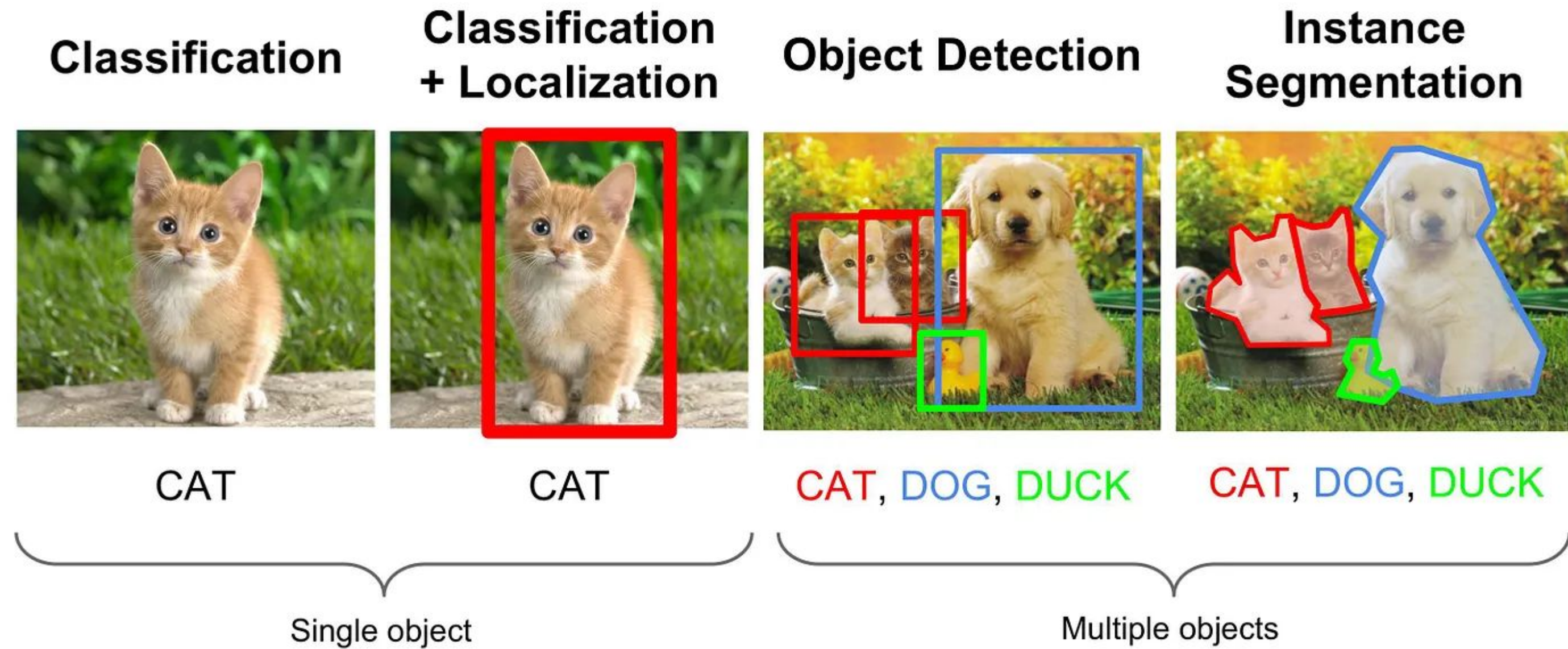


Grayscale Image

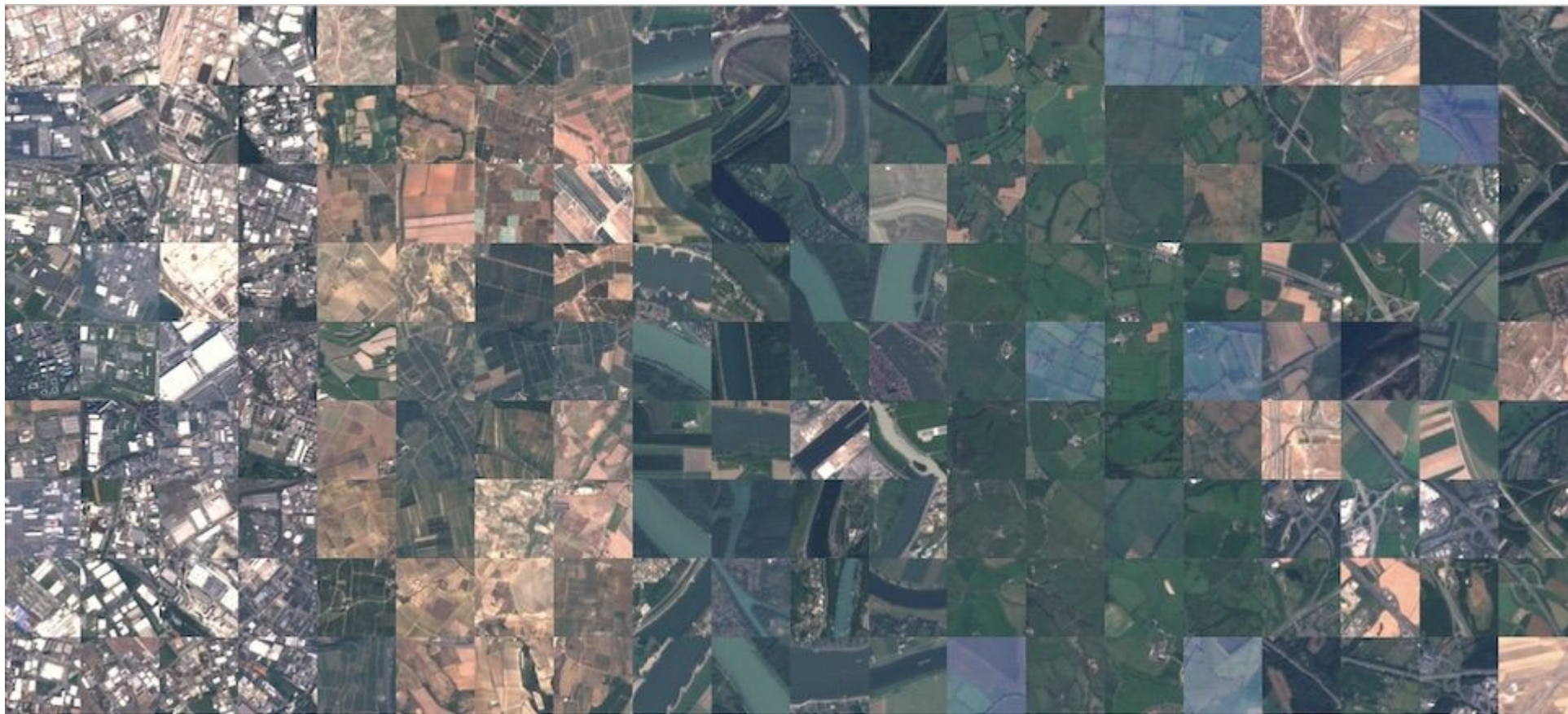


Color Image

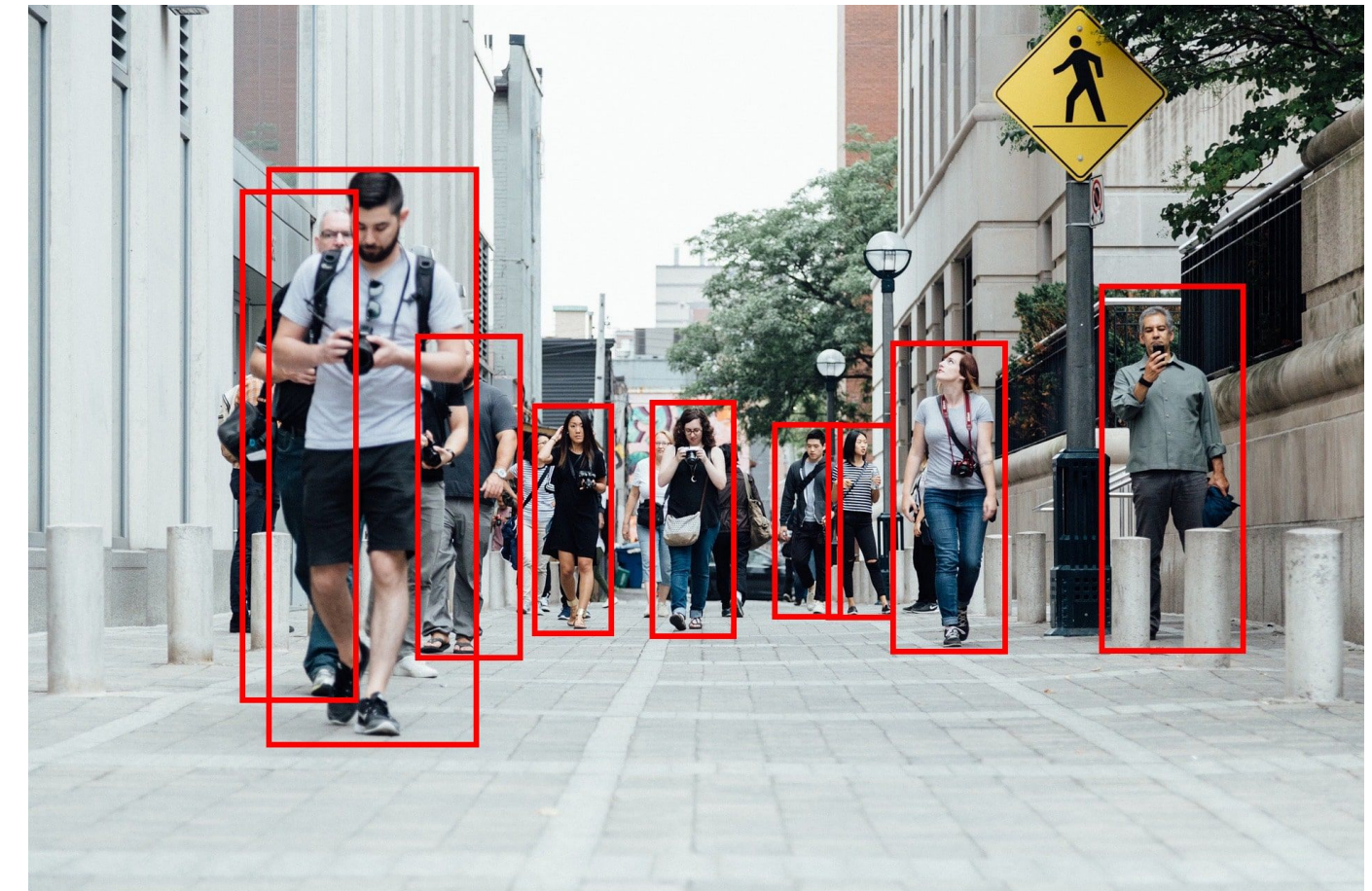
Various Computer Vision tasks



Other examples



Classification: Land Cover Classification Dataset
with classes like Forest area, River, Highway etc ..



Object Detection: People Detection

Other examples ..



Segmentation: Lungs Segmentation



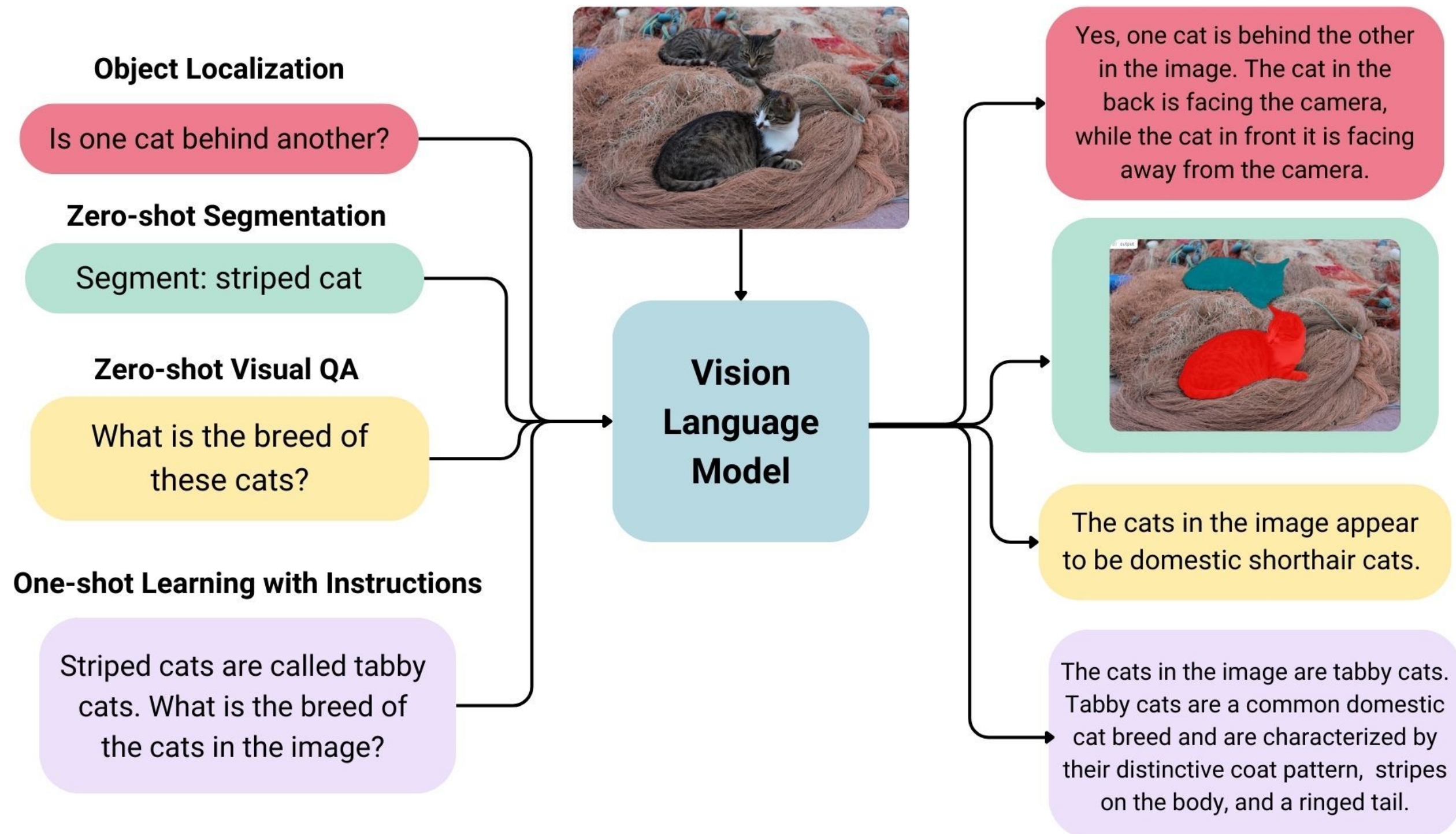
Instance Segmentation: cars, road, pedestrian, cyclists etc ..

Section 02

Vision Language Models

Section 2

Vision Language Models (VLMs)



Vision language models are broadly defined as multimodal models that can learn from images and text.

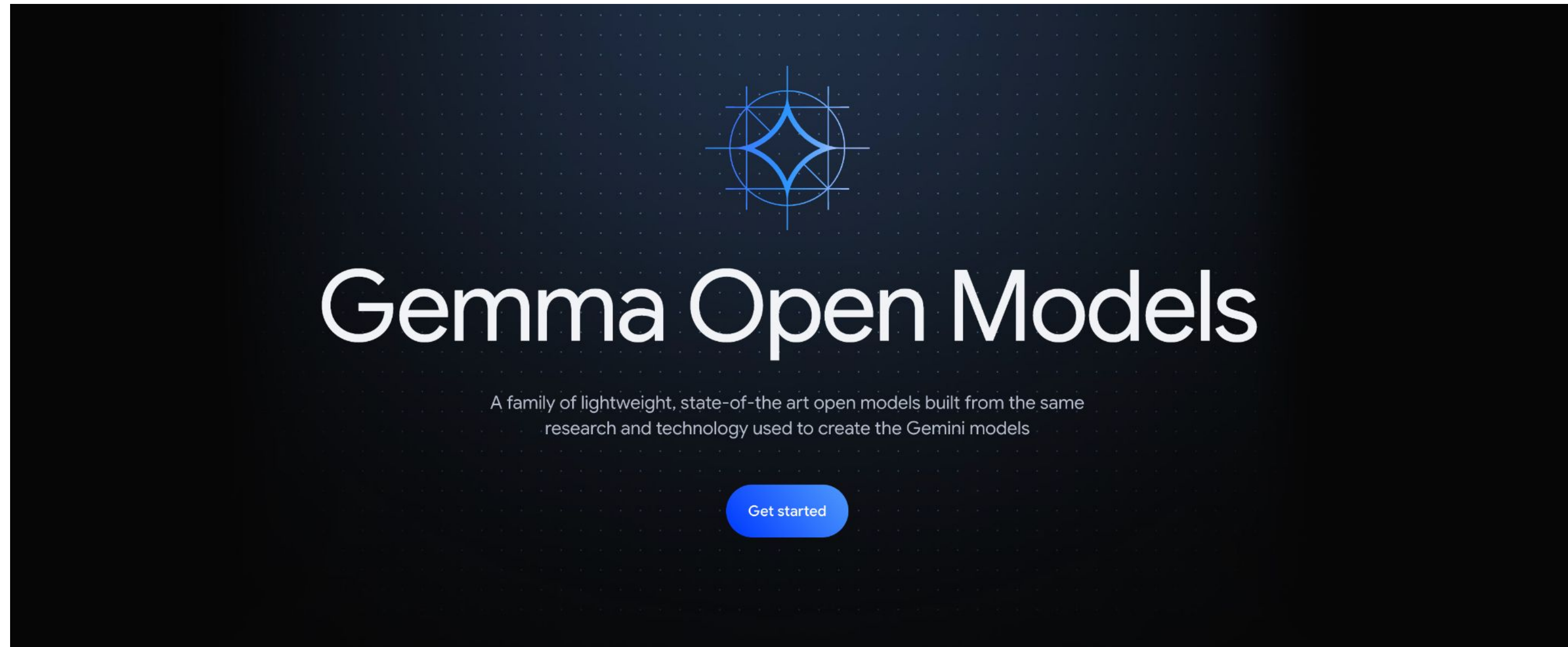
Type of generative models that take image and text inputs, and generate text outputs.

Section 03

Paligemma

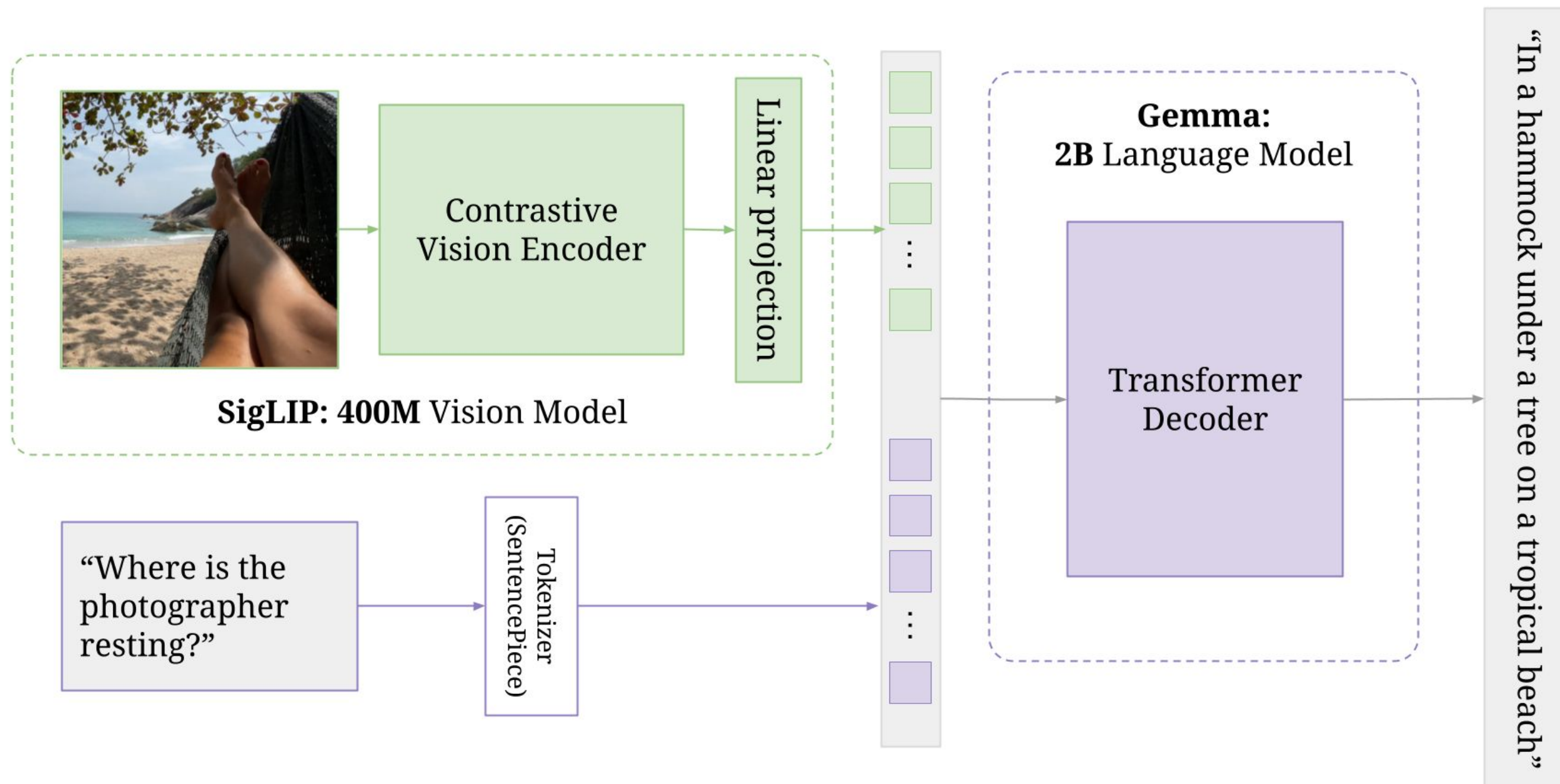
PaliGemma – Google's Cutting-Edge Open Vision Language Model

Paligemma



Paligemma, open vision-language model inspired by PaLI-3, leveraging SigLIP and Gemma, designed as a versatile model for transfer to a wide range of vision-language tasks.

Paligemma Architecture



Types of models

PT checkpoints : pre-trained models that can be fine-tuned to downstream tasks

Ex: paligemma-3b-pt-224, paligemma-3b-pt-448 etc ..

FT checkpoints : set of fine-tuned models, each one specialized to perform a specific task

Ex: paligemma-3b-ft-224, paligemma-3b-ft-448 etc ..

Mix checkpoints : PT models fine-tuned to a mixture of tasks.

Ex: paligemma-3b-mix-896 and other smaller variants

Section 3

Model capabilities: object detection

Image



Prompt

detect cat

Model `paligemma-3b-mix-224` – JAX/FLAX PaliGemma 3B weights, finetuned with 224x224 input images and 256 token input/output text sequences on a mixture of downstream academic datasets. The models are available in float32, bfloat16 and float16 format for research purposes only.

Model

paligemma-3b-mix-224

Decoding

greedy

Run

Clear

Output

<loc0393><loc0630><loc0699><loc0767> cat CAT

Section 3

Model capabilities: object segmentation

Image



Prompt

segment person

Model `paligemma-3b-mix-224` – JAX/FLAX PaliGemma 3B weights, finetuned with 224x224 input images and 256 token input/output text sequences on a mixture of downstream academic datasets. The models are available in float32, bfloat16 and float16 format for research purposes only.

Model

paligemma-3b-mix-224

Decoding

greedy

Run

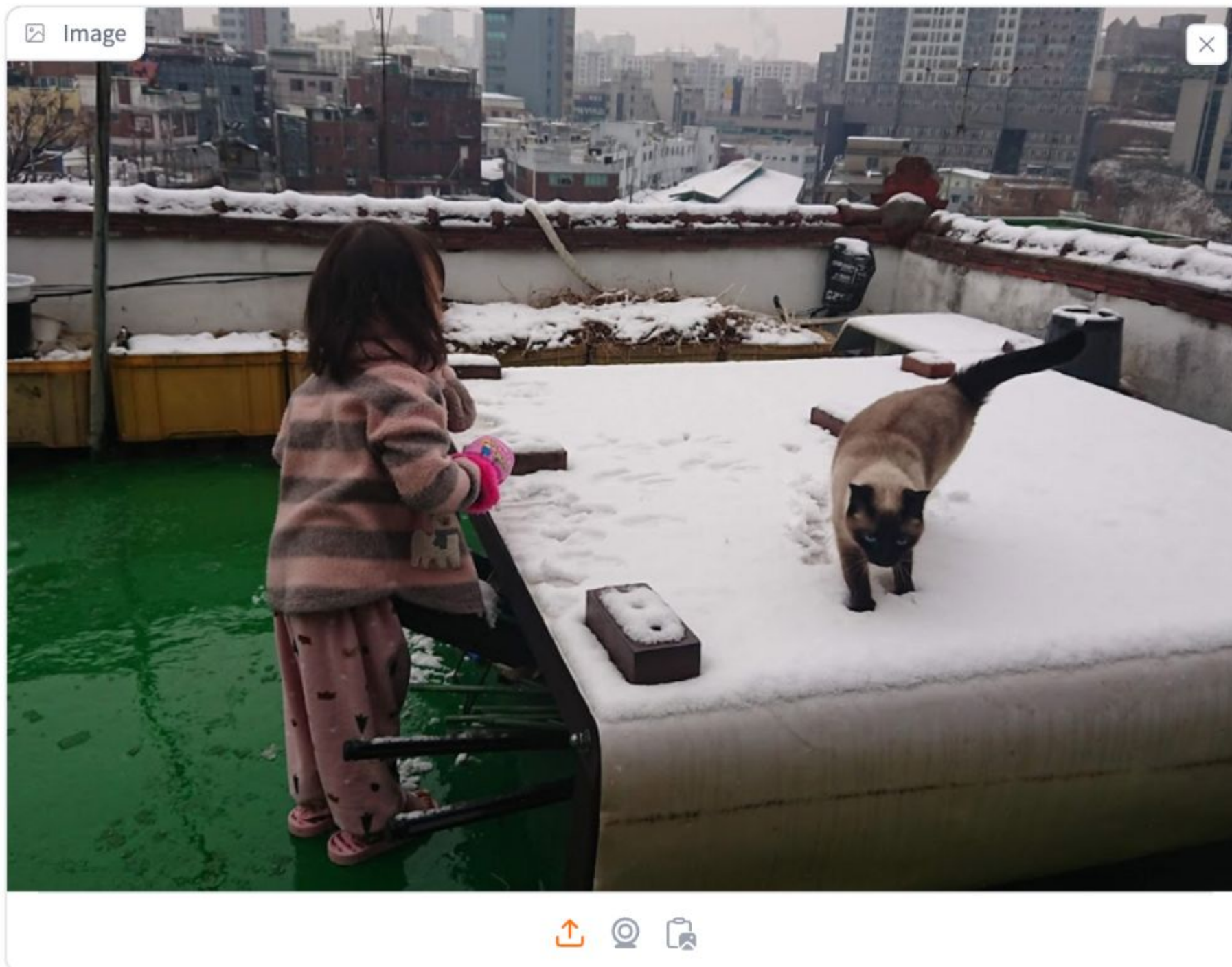
Clear

Output

<loc0246><loc0279><loc0984><loc0421><seg056><seg011><seg057><seg088><seg120><seg026>
<seg052><seg065><seg007><seg035><seg063><seg077><seg049><seg098><seg012><seg089>

Section 3

Model capabilities: image captioning



Prompt

caption the image

Model `paligemma-3b-mix-224` – JAX/FLAX PaliGemma 3B weights, finetuned with 224x224 input images and 256 token input/output text sequences on a mixture of downstream academic datasets. The models are available in float32, bfloat16 and float16 format for research purposes only.

Model

paligemma-3b-mix-224

Decoding

greedy

Run

Clear

Output

In this image we can see a girl standing on the floor. We can also see a cat on the snow, a box, a pole, a wall, a building, a tree and the sky.

Section 3

Model capabilities: OCR



Prompt

ocr

Model paligemma-3b-mix-448 – JAX/FLAX PaliGemma 3B weights, finetuned with 448x448 input images and 512 token input/output text sequences on a mixture of downstream academic datasets. The models are available in float32, bfloat16 and float16 format for research purposes only.

Model

paligemma-3b-mix-448

Decoding

greedy

Run

Clear

Output

BURGERS

All burgers are topped with lettuce, tomato, onion, pickles and yellow American cheese, with mayo and ketchup on the side. Add a side of french fries 2.49

Cheeseburger 8.99

Crispy Chicken 8.99

Bacon Cheeseburger 9.99

WRAPS

All wraps come with lettuce, tomato and a pickle spear on the side 9.99

Section 3

Model capabilities: visual question answering



Prompt

whats the price for cheeseburger ?

Model paligemma-3b-mix-448 – JAX/FLAX PaliGemma 3B weights, finetuned with 448x448 input images and 512 token input/output text sequences on a mixture of downstream academic datasets. The models are available in float32, bfloat16 and float16 format for research purposes only.

Model paligemma-3b-mix-448 Decoding greedy

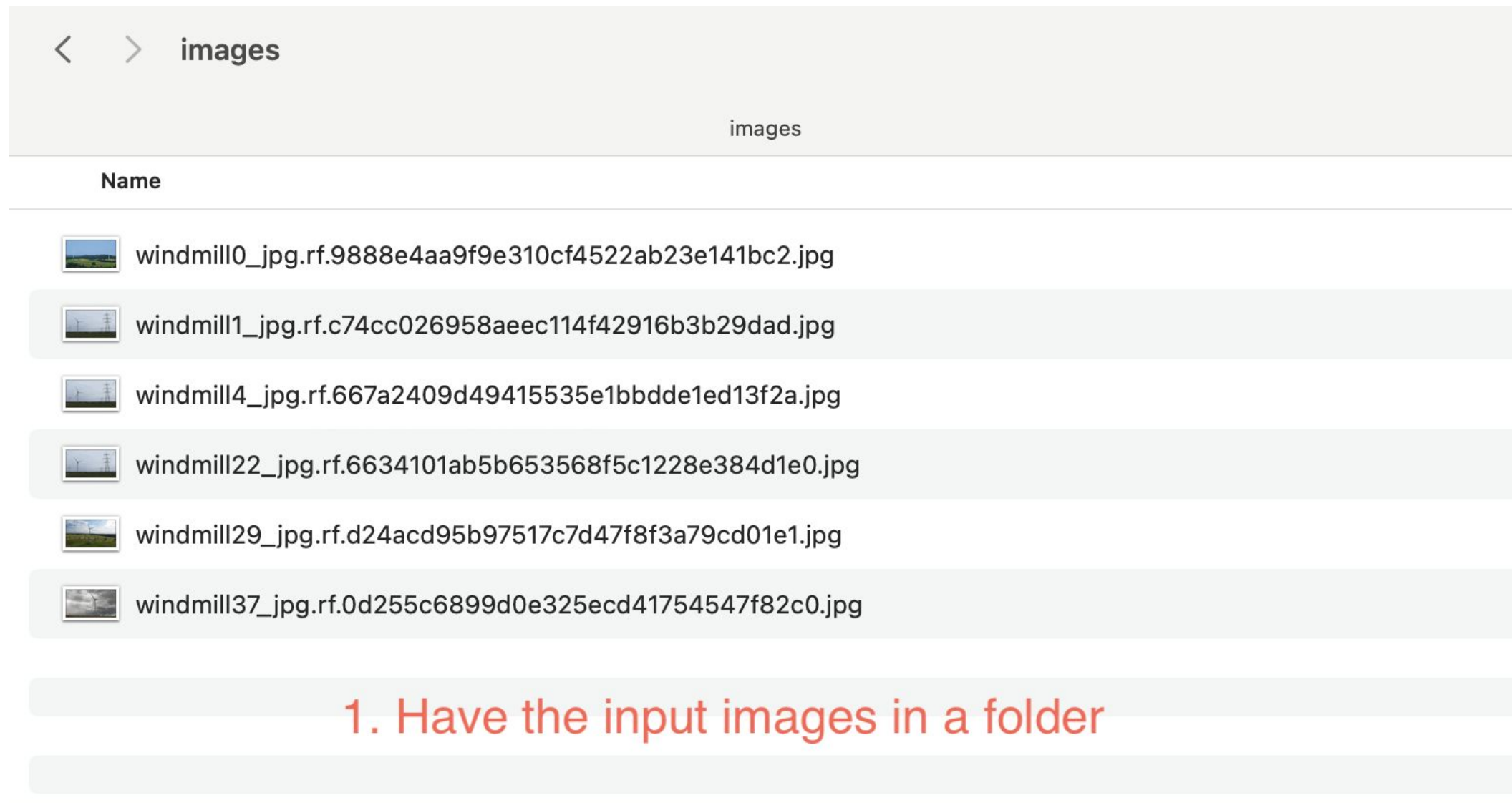
Run Clear

Output

8.99

Section 4

Paligemma for auto labelling



Section 04

Paligemma for auto-labelling codebase



https://github.com/rajesh-bhat/auto_labelling_with_vlms

Thank You !



Rajesh Shreedhar Bhat

Google Developer Expert ML

